◀ FOLD 2: FOLD RIGHT BEHIND LEFT (01 ON COVER) | UNFOLD STEP 1: OPEN RIGHT ▶

PROBABILITY DICE & MODULAR SPINNER PACK

5 Symmetrical Platonic Dice + 10-Sector Decimal Spinner Lid

WHAT IS THIS BOX FOR? OUTSMART YOUR "GUT FEELING."

Human brains are terrible at guessing odds. Five Platonic solids and a 10-sector rotary spinner let you test probability right on your desk.

WELCOME TO YOUR POCKET PROBABILITY LAB

Why guess the odds when you can roll and spin them for real?

- **5 Platonic Solids: 4, 6, 8, 12, and 20-sided symmetrical dice**
- **Free-Spinning Lid Rotor: Flick the pointer for an 8-second spin**
- **10-Sector Decimal Dial: Turns fractions into 10% decimal slices**
- Symmetrical Balance: Every face has an equal shot at landing up

Try this right now: Snap the red pointer onto the lid and give it a flick!

THE LID THAT DOUBLES AS A SPINNER

The storage lid has a built-in 0–9 decimal dial. Snap the pointer into the center hub to run random trials, or pop it inside the case when you pack up.

WHAT'S IN YOUR BOX (9 PIECES)

- 5 Polyhedral Dice (d4 Tetrahedron, d6 Cube, d8 Octahedron, d12, d20)
- 1x Balanced Spinner Pointer + 1x Smooth-Spin Center Hub Pin
- 1x 1x1 Pocket Storage Case + 1x 10-Sector Decimal Dial Lid

★ YOUR FIRST 60 SECONDS WITH THIS BOX START HERE!

1. SNAP THE SPINNER POINTER ONTO THE LID:
Seat the hub pin and pointer into the 0–9 dial lid and give it a flick–each sector is 10%!

2. ROLL ALL 5 PLATONIC DICE AT ONCE:
Compare the 4-sided pyramid (d4 = 25% per face) up to the 20-sided icosahedron (d20 = 5% per face).

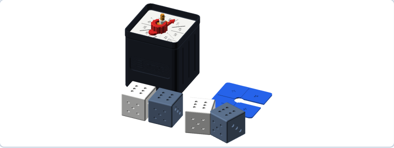
3. PRESS THE BOX ICON KEY [🎲] & [Σ+]:
Press 6 [🎲] on your SC-6 to roll digitally, or log physical rolls with [Σ+] and [g][x̄]!

BOX ICON = CALCULATOR [🎲] DICE SIM, [Σ+] & [G][X̄] STATS KEYS

The button icon embossed on your box lid (left) is the exact calculator key ([🎲], [Σ+] & [g][x̄])! Build every lesson with your blocks, then press the key.

d20 PRIME ODDS (8/20 = 2/5) ↔ BOX [🎲] & [FDISP] KEYS

STEP A: HANDS-ON WITH d20 & POLYHEDRA 1:1 CAD SCENE

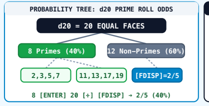


d20 = 20 Faces	Exact Fraction	Box Icon Key
8 Primes (2..19)	8/20 = 2/5 = 40%	[🎲] & [Σ+] Stats

▲ **Count the 8 prime numbers on the 20-sided icosahedron**
(2,3,5,7,11,13,17,19) and reduce 8/20 → 2/5 with [FDISP]!

STEP B: TRY IT ON YOUR CALCULATOR BOX ICON: [🎲], [Σ+] & [g][x̄

PROBABILITY TREE: d20 PRIME ROLL ODDS



8 [ENTER] 20 [÷] [FDISP] → 2/5 (40%)

PRESS BOX ICON & RPN KEYS (SC6):

1. d20 PRIMES:
8 ENTER 20 ÷ = 0.4

2. REDUCE FRAC:
FDISP = 2/5 (40%)

✓ DISPLAY (X) = 2/5

KEYSTROKES: 8 [ENTER] 20 [÷] [FDISP]
Divide 8 prime faces by 20 total faces on the d20 and press [FDISP] to see 2/5!

STEP 1: UNFOLD RIGHT → PAGE 02

HOW IT WORKS: THE CORE IDEA

Platonic Symmetry & Fair Odds: P(Single Outcome) = 1 / Faces

FAIR ODDS, EXPECTED VALUE & THE LONG RUN

For any symmetrical die or balanced spinner with F equal faces:
P(single outcome = k) = 1 / F
What is the average score you should expect over many rolls (1 to F)?
Expected Value E[X] = (F + 1) / 2
And how widely do the rolls spread out around that average?
Variance Var(X) = (F^2 – 1) / 12
Notice that for a 6-sided die, the average roll is 3.5 – a number that doesn't even exist on the die!

WHY ONLY 5 PLATONIC DICE EXIST IN THE UNIVERSE

Over 2,300 years ago, Euclid proved there are only 5 3D shapes where every face is an identical regular polygon and every corner meets at the exact same angle: 4, 6, 8, 12, and 20 faces. You hold all 5 in this box!

THE LAW OF LARGE NUMBERS

In 5 rolls, anything can happen. In 500 rolls, the average settles right onto E[X]. That's how casinos, insurance companies, and weather models turn wild short-term randomness into rock-solid long-term predictions.

3 CLASSIC MATH TRAPS THIS KIT MAKES IMPOSSIBLE VISUAL PROOF

✗ **TRAP 1 – The Gambler's Fallacy ('7 is due!'):**
✓ **FIX:** Dice and spinners have zero memory–after spinning three 9s in a row, the next spin is still 10%.

✗ **TRAP 2 – Thinking 2 and 7 are equally likely on 2d6:**
✓ **FIX:** Only (1,1) makes 2, whereas 6 different pairs make 7 (16.7% vs 2.8%)!

✗ **TRAP 3 – Expecting 10 spins to hit 0–9 once each:**
✓ **FIX:** Real randomness clumps in small samples; convergence only emerges over dozens of trials.

CALCULATOR CHECK: 8 [ENTER] 20 [÷] [FDISP] → X = 2/5

PLATONIC SOLIDS & SPINNER ODDS TABLE

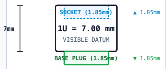
SOLID	FACES (F)	AVG ROLL	SINGLE HIT ODDS
d4 (Tetrahedron)	4 Faces	2.50	1/4 = 25.0%
d6 (Cube)	6 Faces	3.50	1/6 ≈ 16.7%
d8 (Octahedron)	8 Faces	4.50	1/8 = 12.5%
d12 (Dodecahedron)	12 Faces	6.50	1/12 ≈ 8.33%
d20 (Icosahedron)	20 Faces	10.50	1/20 = 5.00%
Spinner Dial	10 Sectors	4.50 (0–9)	1/10 = 10.0%

BUILT FOR FAIR & FAST EXPERIMENTS

- Rounded tumble edges ensure every die rolls smoothly across the desk.
- Low-friction center hub pin lets the pointer spin freely for 8+ seconds.
- Deeply recessed high-contrast numbers won't rub off during classroom play.
- Pocket-sized 1x1 case snaps into any modular StackCalc desk tray.

1:1 CAD REFERENCE VIEW

1:1 BLOCK CAD ANATOMY



✓ FLUSH PLUG-AND-SOCKET JOINT
Socket swallows plug completely:
Only 7.00mm true heights add!

BOX ICON & FORMULA QUICK-REFERENCE

Box Button Icon Keys:
[🎲] Dice Roll, [Σ+] & [g][x̄] Mean

Single Fair Die Law:
P(face) = 1 / F (Sum of all faces = 1.0)

Opposite Faces Rule:
Opposite faces on fair d(F) sum to F + 1

10-Sector Spinner:
1 sector = 1/10 = 0.10 = 10% = 36°

Independent Events:
P(A and B) = P(A) × P(B)

STEP 2: UNFOLD DOWN ▼ PAGES 03–04

▲ FOLD 1: FOLD BOTTOM HALF BEHIND TOP ▲

▼ UNFOLD STEP 2: OPEN DOWN FOR PAGES 03 & 04 ▼

EXAMPLE 1: HANDS-ON WALKTHROUGH

Run 20 Spinner Trials on the 0–9 Decimal Dial Lid

① UNIFORM ODDS: 20 SPINNER TRIALS

1. SNAP ROTARY NEEDLE ONTO DIAL LID:
Seat the pointer arm over the pivot pin on the 10-sector dial lid.
Flick it lightly: verify that it spins freely for at least 8 seconds.

2. CONDUCT 20 TRIAL SPINS:
- Spin the needle 20 times in sequence.
- Tally each result into sectors 0 through 9.

Theoretical Expected Count = 20 × (1/10) = 2 hits per sector.

3. OBSERVE EMPIRICAL VARIANCE:
Some sectors will have 3 or 4 hits; others may have 0 or 1.
→ This is genuine random variance, not equipment bias!

THE LAW OF LARGE NUMBERS IN PRACTICE

Combine tallies across 5 lab partner groups (100 total spins): watch the empirical frequencies level out toward exactly 10% each! Students tangibly experience why large sample sizes produce certainty.

STEP B: TRY EXAMPLE #1 ON YOUR CALCULATOR [🎲], [Σ+] & [g][x̄

STEP 1 (ENTER d6 ODDS):
Press 1 [ENTER] 6 [÷] → pushes P(d6 face) = 0.1667 into Register X.

STEP 2 (MULTIPLY SPINNER):
Press 1 [ENTER] 10 [÷] [×] → multiplies by P(Spinner digit) = 0.10 to get 0.0167.

STEP 3 (EXACT FRACTION):
Press [FDISP] → displays exact compound probability X = 1/60!

LIVE 4-REGISTER RPN STACK READOUT (AFTER KEYSTROKES):
T: 0 | Z: 0 | Y: 0 | X: 1/60 → DISPLAY (X) = 1/60

LESSON 1: 5 PLATONIC DICE & SPINNER ↔ [FDISP] ODDS

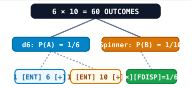
STEP A: HANDS-ON WITH DICE & SPINNER 1:1 CAD SCENE

5 Fair Polyhedra	Decimal Spinner Lid	Compound Odds
d4, d6, d8, d12, d20	10 Sectors (0–9 = 10% e	1/6 × 1/10 = 1/60

▲ **Every Platonic die gives exact fractional odds (1/N), while the**
10-sector lid spinner gives exact decimal odds (0.10 = 10%)!

STEP B: TRY IT ON YOUR CALCULATOR BOX ICON: [🎲], [Σ+] & [g][x̄

INDEPENDENT EVENTS TREE: d6 × SPINNER



P(A and B) = 1/6 × 1/10 = 1/60

PRESS BOX ICON & RPN KEYS (SC6):

1. P(A) & P(B):
1 ENTER 6 ÷ 1 ENTER 10 ÷

2. MULTIPLY:
× FDISP = 1/60

✓ DISPLAY (X) = 1/60

KEYSTROKES: 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP]
Roll the d6 (1/6) and spin the 0–9 lid (1/10), then multiply on the calculator and press [FDISP] for 1/60!
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 1/60 → X = 1/60

CONTINUE RIGHT → PAGE 04

EXAMPLE 2: CORE MATHEMATICAL PRINCIPLE

Two-Dice Sum Bell Curve: Why 7 Appears 6 Times More Than 2

② CENTRAL LIMIT THEOREM: TWO d6 DICE SUM

THE EXPERIMENT: Roll one d12 die vs. rolling two d6 dice.

1. SINGLE d12 DIE (UNIFORM DISTRIBUTION):
Every number from 1 to 12 has exactly 1 face (P = 1/12 ≈ 8.33%).
The histogram is completely flat across all numbers.

2. SUM OF TWO d6 DICE (COMPOUND DISTRIBUTION):
There are 6 × 6 = 36 equally likely microstates (a, b):
- Sum 2: Only 1 way: (1, 1) → P = 1/36 (2.8%)
- Sum 7: Exactly 6 ways: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)

→ P(Sum = 7) = 6/36 = 1/6 (16.7%) – 6x more likely than rolling a 2!

✓ BIRTH OF THE BELL CURVE (CENTRAL LIMIT THEOREM)

Adding independent random variables creates a bell curve!
Uniform + Uniform = Triangular Distribution (Mode = 7)

Students literally watch a smooth bell curve emerge out of discrete square dice rolls. This is the foundational intuition for all **modern statistics and Gaussian normal distributions!**

★ 2-PLAYER GAME: 'PLATONIC HIGH-LOW SHOWDOWN' (2 MINS)

Player A picks the d12 + Spinner (0–9). Player B picks the d20. Before rolling, both players calculate who has the higher expected average (d12 avg 6.5 + Spinner avg 4.5 = 11.0 vs d20 **avg 10.5!), then log 10 rounds into [Σ+] and press [g][x̄]!**

⚙ WHERE THIS SHOWS UP IN QUANTUM PHYSICS, AI & MONTE CARLO SIMS

Why are there only 5 regular polyhedra (d4, d6, d8, d12, d20) in the universe? Euclid proved in 300 BC that no other flat regular polygons can close into a 3D corner! Today, engineers use **random sampling ('Monte Carlo simulation') to predict weather and train AI models.**

LESSON 2: 2d6 EMPIRICAL TRIALS ↔ [Σ+] & [g][x̄] MEAN

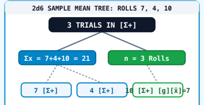
STEP A: HANDS-ON WITH 2d6 BELL CURVE 1:1 CAD SCENE

Sum = 2 or 12	Sum = 7 (Peak)	Log Rolls in [Σ+]
1/36 = 2.8% (Rare)	6/36 = 1/6 = 16.7%	[g][x̄] Reads Mean

▲ **Six different pairs ((1,6), (2,5), (3,4), (4,3), (5,2), (6,1))**
add to 7–log your rolls with [Σ+] and press [g][x̄]!

STEP B: TRY IT ON YOUR CALCULATOR BOX ICON: [🎲], [Σ+] & [g][x̄

2d6 SAMPLE MEAN TREE: ROLLS 7, 4, 10



[g][CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g][x̄] → 7

PRESS BOX ICON & RPN KEYS (SC6):

1. LOG ROLLS:
g CLΣ 7 Σ+ 4 Σ+ 10 Σ+

2. READ MEAN:
g x̄ = 7 (Z: n=3)

✓ DISPLAY (X) = 7

KEYSTROKES: [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄]
Log your physical dice or spinner trials into [Σ+], then press [g][x̄] to see the sample mean converge!
✓ 4-REGISTER STACK: T: 0 | Z: 3 | Y: 0 | X: 7 → X = 7

STEP 3: FLIP SHEET OVER ↻ PAGE 05

◀ PAGE 03 (LEFT) → PAGE 04 (RIGHT) • THEN FLIP SHEET OVER FOR PAGES 05–08 ▶

EXAMPLE 3 & DESK CHALLENGES

Comparing Die Odds vs. Spinner Odds + 3 Desk Challenges

05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • FLIP SIDE (TOP-LEFT)

③ HEAD-TO-HEAD: ROLL THE D20 VS. SPIN THE 0–9 DIAL

THE GOAL: Compare a 20-sided die against the 10-sector decimal spinner!

1. ROLL THE 20-SIDED DIE (d20):
Each face from 1 to 20 has a 1/20 = 5% chance of landing on top.

2. GROUP INTO EVEN VS. ODD:
How many faces are > 18? Only two (19 and 20): 2/20 = 1/10 = 10%.

3. TEST ON THE 0–9 SPINNER:
Spinning a '9' on the 10-sector lid also has a 1/10 = 10% chance—identical odds!

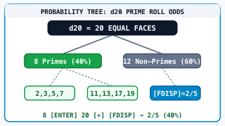
✓ TRANSLATING FRACTIONS INTO DECIMALS BY SPINNING

Dice naturally teach fractions: 1/4, 1/6, 1/8, 1/12, and 1/20.
The 10-sector spinner naturally teaches decimals and percents: 1 sector = 0.10 = 10%.
Using both together shows that a 1-in-4 roll on the d4 (25%) is halfway between 2 sectors (20%) and 3 sectors (30%) on the decimal spinner!

PRO TIP: CLOSE YOUR EYES & FEEL THE ANSWER

Once you build an example, run your fingers across the pieces. Your hands remember physical shapes and heights way longer than memorized rules!

EXAMPLE 3: VISUAL DECOMPOSITION TREE



8 [ENTER] 20 [÷] [FDISP] → 2/5 (40%)

STEP B: PRESS BOX ICON & RPN KEYS (SC6)

TACTILE KEYSTROKE WALKTHROUGH:

1. d20 PRIMES:
8 ENTER 20 ÷ = 0.4

2. REDUCE FRAC:
FDISP = 2/5 (40%)

✓ CALCULATOR DISPLAY (X) = 2/5

FULL KEYSTROKES: 8 [ENTER] 20 [÷] [FDISP]
✓ 4-REGISTER STACK: T: 0 | Z: 0 | Y: 0 | X: 2/5 → X = 2/5

3 DESK CHALLENGES – TRY THESE NOW!

WARM-UP #1 **Spin the 0–9 dial 30 times. Expected hits per number = 3. Which won?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #1 below!**

LEVEL-UP #2 **Roll the d12 and d20 together. Which die is more likely to roll a 1?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #2 below!**

BRAIN-MELTER #3 **Roll the d20: what is the exact probability of rolling a prime number?**
Step A: Build with blocks on your Lid Tray • Step B: Verify on SC-32 Calculator
✓ **Build first, then flip booklet upside-down to check Answer #3 below!**

SELF-CHECK ANSWER KEY (ROTATE BOOKLET 180° TO READ)
#1: Real randomness clusters! | #2: d12 (1/12 = 8.3% vs 1/20 = 5%)
#3: 8 primes / 20 = 40%!

FOR PARENTS & TEACHERS (Grades 5–10 (Probability, Statistics & Data Science))

- CCSS.MATH.CONTENT.7.SP.C.5: Understand that probability is a number between 0 and 1.
- CCSS.MATH.CONTENT.7.SP.C.6: Approximate probability by collecting data on chance processes.
- CCSS.MATH.CONTENT.7.SP.C.8: Find probabilities of compound events using tables and lists.
- **CCSS.MATH.PRACTICE.MP2, MP4, MP5, MP7: Reason, Model, Use Tools & Structure.**

Next up on Page 06 → How to stack everything back in the box in 10 seconds!

BOSS CHALLENGE #4 **BLOCKS → BOX ICON KEY SPEED-RUN!**

1. Build Examples #1, #2, and #3 hands-on with your physical pieces.
2. Press the Box Button Icon ([🎲], [Σ+] & [g][x̄]) and keystrokes on your calculator:

- 8 [ENTER] 20 [÷] [FDISP] → X = 2/5

CHECK OFF YOUR MASTERY MILESTONES:

☐ 1. Demonstrated Lessons #1, #2 & #3 hands-on with the physical blocks
☐ 2. Solved all 3 Desk Challenges above (check upside-down key if stuck!)
☐ 3. Demonstrated each lesson on the calculator using the Box Button Icon
☐ 4. Packed all pieces back into the box in under 10 seconds (see Page 06 →)

TEST YOURSELF • ANSWERS INCLUDED SO YOU CAN CHECK RIGHT AWAY

NEXT: HOW TO PACK UP → PAGE 06

HOW TO STACK IT ALL BACK IN

The 10-Second Self-Checking Pack-Up Puzzle

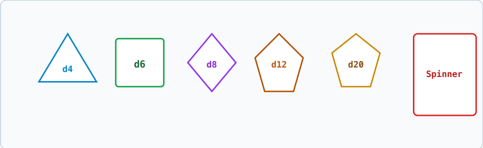
05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • PACK-UP GUIDE (TOP-RIGHT)

TOP-DOWN BOX MAP: WHERE EVERY PIECE LIVES

Putting the blocks away is a 10-second puzzle. Match each piece to its labeled home bay below so everything nests flush and the lid snaps shut:

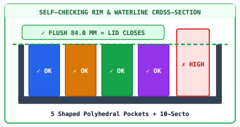


✓ CUSTOM CAVITIES FOR ALL 5 PLATONIC SOLIDS

Each of the 5 polyhedral dice (d4, d6, d8, d12, d20) drops into its own shaped pocket inside the 1x1 storage case, alongside the spinner pointer and pin. Before snapping the dial lid shut, a 2-second look confirms all 5 dice are in place and none rolled under the couch!

1. STORAGE CASE & COMPARTMENTS

2. WHY THE LID SELF-CHECKS EVERY PIECE


SELF-CHECKING RIM & WATERLINE CROSS-SECTION
✓ FLUSH 84.0 MM = LID CLOSES
✓ OK
✓ OK
✓ OK
✓ OK
✗ HIGH
5 Shaped Polyhedral Pockets + 10-Secto

▲ SELF-INDEXING BAYS & EMBOSSED BOX BUTTON ICON:
Calibrated bays guide every piece home, and the lid icon links directly to your calculator.

THE 10-SECOND PACK-UP ROUTINE

HOW TO STACK EVERYTHING BACK IN:

1. SORT BY COLOR & NUMBER:
Group matching blocks or tiles together on your desk first.

2. STACK & DROP INTO LABELED BAYS:
Follow the Top-Down Box Map on the left. If a tower looks too tall or too short, a block is in the wrong bay—swap it until everything levels!

3. TUCK THIS CARD IN THE LID & SNAP SHUT:
Fold this guide back to [01] and rest it right on top before closing.

10-SECTOR SPINNER LID & POINTER STORAGE

- Snap the pointer onto the lid's center hub to spin, or tuck it inside the box for travel.
- Pocket-sized 1x1 case docks neatly beside your SC-32 calculator or larger kit boxes.
- Keep away from toddlers under 3 (polyhedral dice are small parts).

SHORT A PIECE AT CLEAN-UP TIME?

First check under notebooks and chair legs! If a piece truly vanished, **see PAGE 08 (bottom-right) for your complete Part ID Reference Table.**

✓ PROB-01 4-POINT CLEAN-UP CHECKLIST

1. **SEAT 5 POLYHEDRA:** Drop d4, d6, d8, d12, and d20 into their shaped pockets inside the 1×1 case.
2. **TUCK POINTER & PIN:** Store the Blue spinner pointer and Clear hub pin in the side bay.
3. **FOLD CARD TO [01]:** Fold this reference booklet to Panel [01] for compact storage.
4. **SNAP SPINNER LID:** Click the 10-sector decimal dial lid onto the 1×1 case!

MODULAR CLASSROOM & BOOKSHELF NESTING (42.0 MM PITCH)
Every StackCalc box is sized in exact multiples of 42.0 mm (1×1, 3×1, 3×2).
All 9 kit boxes interlock side-by-side in a classroom cart or drawer with zero wasted space, and every embossed box icon matches its calculator operation key!

SHAPED DIE CAVITIES • EVERY POLYHEDRON HAS ITS OWN HOME SOCKET

NEXT: CONNECTED LAB ▼ PAGE 07

SIDE B (PAGES 05–08) • READ TOP ROW FIRST

THEN READ BOTTOM ROW (PAGE 07 → PAGE 08)

THE STACKCALC CONNECTED LAB

How Your Kits & RPN Calculator Stack Work Together

05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • ECOSYSTEM (BOT-LEFT)

ONE CONNECTED SYSTEM FROM KINDERGARTEN TO CALCULUS

Every StackCalc box uses the same modular grid and unit heights—so pieces from different kits actually fit and work together.

WHERE YOUR BOX FITS IN THE 9-KIT LAB

BASE-10 **Ten-Frames Lab** Early Counting & Regrouping	DOMI-01 **Math Domino Studio** Doubles & 12-Frame Part-Whole	FRAC-01 **Fraction Studio** Proportional Towers & 4-Way Faces
PRIME-01 **Prime Factor Studio** Multiply by Stacking Prime Towers	ALGB-01 **Algebra Studio** Quadratic Area Arrays & Factoring	TIME-01 **Planetary Clock** 12:1 Geared Hands & Time Wedges
PROB-01 ◀ YOUR BOX **Probability Dice** Fair Polyhedrals & 10-Sector Spinner	VOLV-01 **Unit Circle Volvelle** Spin-to-Read Exact Trig Radicals	SC-32-HW **SC-32 RPN Calculator** 4-Register Stack Scientific Engine

WHY THIS MATTERS: ZERO "THROW-AWAY" MATH

In most classrooms, every grade uses a totally unrelated plastic toy.
In StackCalc, the 7mm unit block you count with in 1st grade is the exact same height as the 1/12th fraction block and the [2] prime block later on!

1. PROB-01 BOX LID ICON

[🎲], [Σ+] & [g][x̄]
EMBOSSED ON BOX LID
Build First → Press Key!

2. SC-6 ELEMENTARY RPN

3. SC-32 SCIENTIFIC RPN

▲ BOX ICON ↔ CALCULATOR KEY CONTINUITY:
Every physical box lid carries the exact icon of the calculator key that computes that lesson!

HOW BLOCKS CONNECT TO THE CALCULATOR

SHARED MODULAR FOOTPRINT & HEIGHTS

- The compact 1x1 case shares the exact modular grid footprint of the rest of the StackCalc lab.
- Recessed sockets swallow the bottom peg so only true heights add.
- All storage boxes nest neatly side-by-side on any desk or shelf.

WHY WE CALL IT "STACKCALC" (THE RPN SECRET)

- Enter your experimental dice tallies into the SC-32 calculator to compute live means and sums!
- Placing a block on the table = pressing [ENTER] to push onto the stack.
- Snapping two towers together = pressing [+] or [×] with zero parentheses!
- Mirrors the 4 stack levels on the SC-32 calculator: X, Y, Z, and T.

FREE ACTIVITY DECKS & INTERACTIVE SIMULATORS

Download extra printable puzzle cards, teacher lesson guides, and interactive browser simulations anytime—no login needed.

Explore All 9 Kits & Free Guides at stackcalc.io

LIVE 4-REGISTER RPN STACK

T: 0 (Top Memory)
Z: 0 (3rd Level)
Y: 0 (1st Block)
X: 1/60 (P(A)×P(B)) (Active Display)

HOW BLOCKS MAP TO STACK KEYS

1. PLACE 1ST PIECE ON STAGE:
Type number + [ENTER] → lifts to Y

2. ADD / STACK 2ND PIECE:
Type 2nd number in X, then [+]/[×]

3. PRESS BOX ICON KEY:
Press [🎲], [Σ+] & [g][x̄]
✓ **Zero Parentheses • Always Visible!**

BOX ICON = [🎲], [Σ+] & [g][x̄] • VERIFIED LESSON SEQUENCES:
- Lesson 1: 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDIS → X = 1/60
- Lesson 2: [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → X = 7

MODULAR 9-KIT CONTINUUM • FROM EARLY COUNTING TO AP CALCULUS

NEXT: PART ID TABLE → PAGE 08

KEEP THIS CARD! PART IDs & VIDEO

Complete 9-Piece Reference & 60-Second 3D Video Guide

05 06 07 08

SKU: PROB-01 • KEEP IN LID

SIDE B • PART REFERENCE (BOT-RIGHT)

PART ID REFERENCE TABLE (9 TOTAL PIECES)

✓ KEEP THIS CARD INSIDE YOUR BOX LID!

Backpacks eat blocks. Dogs chew dice. Vacuum cleaners take no prisoners. Keep this card in your lid so you always have the exact Part IDs below for single-piece refills!

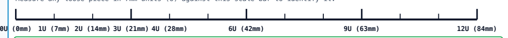
PART NAME	QTY	COLOR / FINISH	PART ID
Tetrahedron Die (d4)	1x	Black / Gold	PROB-DICE-04a
6-Sided Cube Die (d6)	1x	Black / Gold	PROB-DICE-04a
Octahedron Die (d8)	1x	Black / Gold	PROB-DICE-04a
Dodecahedron Die (d12)	1x	Black / Gold	PROB-DICE-04a
Icosahedron Die (d20)	1x	Black / Gold	PROB-DICE-04a
Balanced Spinner Pointer	1x	Blue	PROB-SPIN-7e5
Smooth-Spin Hub Pin	1x	Clear	PROB-SPIN-7e5
1x1 Case & Dial Lid	2x	Black	PROB-DICE-04a

NEED A REPLACEMENT PIECE? NO NEED TO BUY A WHOLE NEW BOX.
Enter any Part ID above at **stackcalc.io/parts** to order individual replacement pieces.

COLOR & PIECE IDENTIFICATION KEY MATCH ANY LOOSE PART

Black / Gold	Blue	Clear	Black
Platonic Dice (d4..d20)	Balanced Spinner Pointer	Smooth-Spin Hub Pin	1x1 Case & Dial Lid

UNIVERSAL STACKCALC HEIGHT PITCH (1U = 7.00 MM • 12U = 84.00 MM ONE WHOLE)
Measure any loose piece in 7mm units (U) against this scale bar to identify it:



✓ DISTRICT PO & SAFETY: 10-Pack SKU PROB-01-CLASS10 • CPSIA / ASTM F963 Non-Toxic Washable PLA/ABS

WATCH THE 60-SECOND 3D VIDEO GUIDE

POINT CAMERA TO WATCH DEMO
stackcalc.io/products/dice-and-spinner/
See animated 3D examples & pack-up walkthrough

QUESTIONS, CLASSROOM SETS & SUPPORT

- Email us anytime: support@stackcalc.io
- School & District POs: school-orders@stackcalc.io

Designed in Mountain View, CA • Non-Toxic & Washable

COMPANION MANUAL CHAPTERS, STANDARDS & KEYSTROKES

BOOK CHAPTERS: SC-6 Guide Ch. 6 (Probability & Stats) • SC-32 Manual Ch. 9
STANDARDS: CCSS 6.SP.A.1, 7.SP.C.5, 7.SP.C.8 • TEKS 7.6E, 7.6I (Sample Space & Mean)
WATERLINE SPEC: 5 Shaped Polyhedral Pockets + 10-Sector Spinner Lid

ALL 3 VERIFIED CALCULATOR LESSON SEQUENCES (SC6):
Ex 1: 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → X=1/60
Ex 2: [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → X=7
Ex 3: 8 [ENTER] 20 [÷] [FDISP] → X=2/5

BOX ICON KEY: [🎲], [Σ+] & [g][x̄]
✓ Cross-Referenced to Companion Book & Calculator Engine
Fold back to [01] & keep this reference card inside your box lid!

LOST A PIECE? GRAB SINGLE REPLACEMENTS ANYTIME AT stackcalc.io/parts

✓ KEEP THIS GUIDE IN BOX LID!